Lec 1 - Review 1

function: A function (abbreviated fun) is a rule that assigns, for each element in one set (called the domain) exactly one element from another set (called the codomain).

Note: Codomain is often mistaken for "range", sometimes they are defined, but in this course "range" is acceptable.

$y = f(x)$ — dependent variable; independent variable; name of the function (can be more complicated than you think)

Even function $\rightarrow f(x) = f(-x)$
Odd function $\rightarrow -f(x) = f(-x)$

HLT
↳ Let f be a function. If f never assigns the same y-value to two different x-values, then f is one-to-one.

Quadratic formula:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

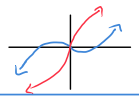
discriminant: $D = b^2 - 4ac$

- $+ \rightarrow$ 2 real roots
- $0 \rightarrow$ 1 real root
- $- \rightarrow$ no real roots

If f passes HLT, f has an inverse, called f^{-1}.

$f^{-1}(y) = x \iff y = f(x)$

Note: The graph of $y = f^{-1}(x)$ is just the graph $y = f(x)$ reflected across the diagonal



Lec 2 - Review 2

Exponential function

$y = C^x$

Laws of exponent

$a^0 = 1, \ a \neq 0$

$a^1 = a$

$a^{x+y} = a^x \cdot a^y$

$(a^x)^y = a^{x \cdot y}$

$(ab)^x = a^x \cdot b^x$

$a^{-x} = \frac{1}{a^x}$

Logarithmic function

$y = \log_c x$

$y = c^x \longleftrightarrow x = \log_c y$

laws of logarithms

$\log_a(1) = 0$

$\log_a(a) = 1$

$\log_a(xy) = \log_a x + \log_a y$

$\log_a(x^p) = p \log_a x$

$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$

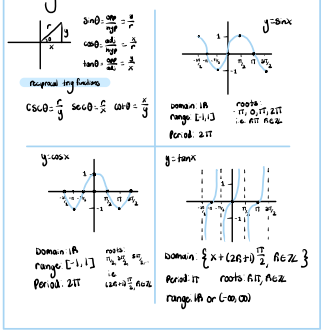
$\log_a(x) = \frac{\ln x}{\ln a}, \ a > 0, a \neq 1, x > 0$

Trigonometric functions

$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{y}{r}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{x}{r}$, $\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{y}{x}$

reciprocal trig functions

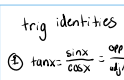
$\csc\theta = \frac{r}{y}$, $\sec\theta = \frac{r}{x}$, $\cot\theta = \frac{x}{y}$



trig identities

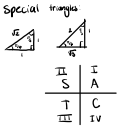
① $\tan x = \frac{\sin x}{\cos x} = \frac{\text{opp/hyp}}{\text{adj/hyp}} = \frac{\text{opp}}{\text{adj}}$

② $\sin^2\theta + \cos^2\theta = 1$



$\tan^2\theta + 1 = \sec^2\theta$

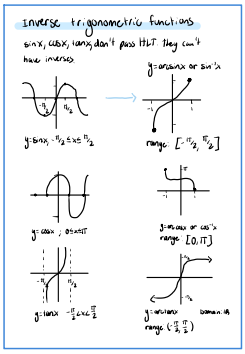
Special triangles



II	I
S	A
T	C
III	IV

Inverse trigonometric functions

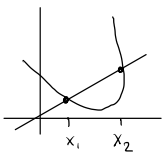
$\sin x$, $\cos x$, $\tan x$ don't pass HLT, they can't have inverses



Math 1320

Lec 3 - Limits & indeterminate forms

(AROC, secant lines): If a fun f is defined at x_1 and x_2, then the average rate of change (AROC) of f between x_1 and x_2 is the same as the slope of the secant line from x_1 to x_2.



$$\text{AROC} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

Limits: Suppose $f(x)$ is defined for all x "near" a. Then $\lim_{x \to a} f(x) = L$ (pronounced "the limit of $f(x)$ as x approaches a is L") if $f(x)$ can get as close to L as we like as long as a is close enough to a (but not $x = a$).

One-sided limits: We write $\lim_{x \to a^-} f(x) = L$ ("The limit of $f(x)$ as x approaches a from the left is L") if $f(x)$ gets as close to L as we choose $x < a$ close enough to a. $\lim_{x \to a^+} f(x) = L$ is defined similarity.

IMPORTANT: $\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} = L = \lim_{x \to a^+} f(x)$

Note: If we are dealing with a "nice" fun (no jumps, no holes, no vertical asymptotes), then we evaluate a limit as $x \to a$ by just subbing $x = a$. often we need to simplify before we get to that point

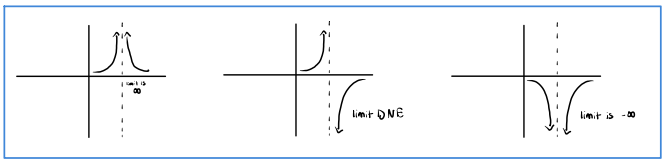
Lec 4 - Continuity & Limits to infinity

Continuous: A fun is continuous (cts) at a point "a" if $\lim_{x \to a} f(x)$ exists and is equal to $f(a)$. We say f is cts on its domain if it's cts at every point in its domain.

Note: Note that continuity at a point "a" requires us to verify $\left(\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)\right)$ That the limit exists. The following funs and their transformations are cts on their domains:
Polynomials, rational funs, $y = |x|$, inverse trig funs, exponential funs, logarithmic funs

Limits equalling ∞: We say $\lim_{x \to a} f(x) = \infty$ (or $-\infty$) to mean that $f(x)$ gets as large as the line for as large and negative as long as we take x close enough to a technically, such a limit DNE (since $\pm\infty$ are not numbers), but if a limit goes to I too, we will always specify this instead of just saying "DNE". If we just say "DNE" that means the one-sided limits disagree.

Note: A limit of this type will usually arise when we attempt to divide a non-zero number by something that is approaching zero.
A fun $f(x)$ has a vertical asymptote (VA) at $x = a$ if and only if $\lim_{x \to a^-} f(x) = \pm\infty$ or $\lim_{x \to a^+} f(x) = \pm\infty$



Evaluating $\lim_{x \to a} f(x)$...
If it's a piece wise fun (including absolute values) where pieces meet up at $x = a$, immediately check one-sided limits.

check "$f(a)$"

- If indeterminate we must simplify further to figure out what's going on
  Indeterminate forms include: $\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, \infty^0, 1^\infty$
  Possible tactics:
  - factor + cancel
  - rationalize + cancel
  - find common denominator once simplified, return to step
- If $f(a)$ is a number, this is probably the value of the limit (need f cts at a)
- If you see $\frac{c}{0}$ for some $c \neq 0$ then limit DNE now you must determine the sign of the one-sided limits, which are usually each $= \infty$

(limits as $x \to \infty$): Suppose f is defined for all $x > a$ for some ..... we say $\lim_{x \to \infty} f(x) = L$ to mean that $f(x)$ gets as close to L as we like as long as x is enough $\lim_{x \to -\infty} f(x) = L$ is defined similarity (L could also be $\pm\infty$ as in def 4.2)

with rational functions, we can skip a lot of steps

compare the largest powers of x in the numerator & denominator

- If denom. has higher power, limit is 0.
- If they're equal the limit is equal the ratio of the coefficients
- If num has higher power: limit goes $\pm\infty$ check signs of the coefficients of the highest powers of num & denom.

Lec 5 - Differentiation: Definition & Basic rules

AROC $\rightarrow m = \frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$, $(a, f(a))$ to $(b, f(b))$ $\rightarrow$ secant

Instantaneous rate of change $\rightarrow$ $b = a + h$, $h \to 0$ $\rightarrow$ tangent

(Derivative at a point):

The derivative of f at "a" is $\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$, provided this limit exists. while this expression is called the derivative of f at "a" the act of finding the derivative is called differentiation (or differentiating) not "deriving"!

Differentiable: A fun is called differentiable (diff) at "a" if $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists.

Note: f cannot be diff unless it's cts because that's required for hm $(f(a+h)-f(a))$ to equal 0, which produces the $\frac{0}{0}$ indeterminate. So if f is diff at "a", then it's cts at "a".

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

- f is diff at any point where f' is defined
- f'(x) gives the slope of the tangent of f at x

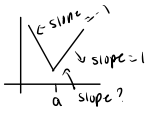
How functions can fail to be differentiable

1) Remember if (f diff at a) then (f cts at a) so if f isn't cts, it isn't diff!

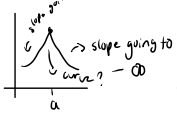
what is the slope of the tangent at a? Non sense!

2) Corners and cusps:

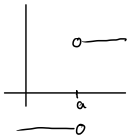
corner



cusp



Graph of derivative

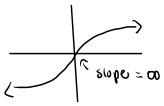


Graph of derivative

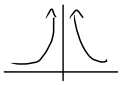
In both cases f' is not cts at a

3)

Vertical tangents:



$y = \sqrt[3]{x}$ is not diff at $x=0$



graph of derivative

Lec 6/7 Differentiation

1) $\frac{d}{dx} c = 0$ if $c \in \mathbb{R}$ is a constant

2) $\frac{d}{dx} cf(x) = cf'(x)$ if $c \in \mathbb{R}$ is a constant

3) $\frac{d}{dx}(f(x)+g(x)) = f'(x) + g'(x)$

4) $\frac{d}{dx} x^n = nx^{n-1}$ for any $n \in \mathbb{R}$

5) $\frac{d}{dx} f(x)g(x) = f'(x)g(x) + f(x)g'(x)$ product rule

6) $\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$ quotient rule

7) $\frac{d}{dx} e^x = e^x$ (e^x rule)

8) $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

9) $\frac{d}{dx} a^x = a^x \cdot \ln a$ if $a>0, a \neq 1$

10) $\frac{d}{dx} \sin x = \cos x$

11) $\frac{d}{dx} \cos x = -\sin x$

12) $\frac{d}{dx} \tan x = \sec^2 x$

13) $\frac{d}{dx} \csc x = -\csc x \cot x$

14) $\frac{d}{dx} \sec x = \sec x \tan x$

15) $\frac{d}{dx} \cot x = -\csc^2 x$

Lec 8 - Differentiation: Inverses & Implicit Differentiation

Ex: Find $\frac{dy}{dx}$, where $5y + 2\cos x = \cos(y+x^2) + \sin y$ (Implicit differentiation)

$$5\cdot\frac{dy}{dx} - 2\sin x = -\sin(y+x^2)\cdot\left(\frac{dy}{dx}+2x\right) + \cos y \cdot \frac{dy}{dx}$$

$$5\cdot\frac{dy}{dx} - 2\sin x = -\sin(y+x^2)\cdot\frac{dy}{dx} - 2x\sin(y+x^2) + \cos y\cdot\frac{dy}{dx}$$

$$5\cdot\frac{dy}{dx} + \sin(y+x^2)\cdot\frac{dy}{dx} - \cos y\cdot\frac{dy}{dx} = -2x\sin(y+x^2) + 2\sin x$$

$$\frac{dy}{dx}(5+\sin(y+x^2) - \cos y) = -2x\sin(y+x^2) + 2\sin x$$

$$\frac{dy}{dx} = \frac{-2x\sin(y+x^2) + 2\sin x}{5+\sin(y+x^2) - \cos y}$$

Recall for inverses: $(f\circ f^{-1})(x) = x$, i.e. $f(f^{-1}(x)) = x$ (differentiating inverses)

Differentiate both sides wrt x

LHS: $f'(f^{-1}(x))\cdot\frac{d}{dx}f^{-1}(x)$ RHS: $= 1$

$$\Rightarrow \frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$$

$f(x) = e^x$ & $f^{-1}(x) = \ln x$

$\frac{d}{dx}\ln x = \frac{1}{e^{\ln x}} = \frac{1}{x}$

or rule: $\frac{d}{dx}\ln x = \frac{1}{x}$

$\frac{d}{dx}\log_a(x) = \frac{1}{a^{\log_a(x)}\ln a} = \frac{1}{x\ln a}$

Ex: $\ln(f(x)) = \ln(x^{x+1})$ differentiation of $f(x) = x^{x+1}$, $x>0$

$= (x+1)\ln x$

$$\frac{1}{f(x)}\cdot f'(x) = 1\cdot\ln x + (x+1)\cdot\frac{1}{x}$$ (logarithmic differentiation)

$$\Rightarrow f'(x) = f(x)\left(\ln x + \frac{x+1}{x}\right)$$

$$= x^{x+1}\left(\ln x + \frac{x+1}{x}\right)$$

$$5x^3y^4 - 2x^2\cos y + 4y^3 = x^2$$

$$5\left(3x^2y^4 + 4x^3y^3\cdot\frac{dy}{dx}\right) - 2\left(2x\cdot\cos y - x^2\sin y\cdot\frac{dy}{dx}\right) + 4\left(3y^2\cdot\frac{dy}{dx}\right) = 2x$$

$$\left(15x^2y^4 + 20x^3y^3\cdot\frac{dy}{dx}\right) + \left(-4x\cos y + 2x^2\sin y\frac{dy}{dx}\right) + \left(12y^2\cdot\frac{dy}{dx}\right) = 2x$$

$$20x^3y^3\cdot\frac{dy}{dx} + 2x^2\sin y\frac{dy}{dx} + 12y^2\cdot\frac{dy}{dx} = 2x + 4x\cos y - 15x^2y^4$$

$$\frac{dy}{dx}(20x^3y^3 + 2x^2\sin y + 12y^2) = 2x + 4x\cos y - 15x^2y^4$$

$$\frac{dy}{dx} = \frac{2x + 4x\cos y - 15x^2y^4}{20x^3y^3 + 2x^2\sin y + 12y^2}$$

$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$

If we let $f(x) = \sin x$, $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$,

and $f^{-1}(x) = \arcsin x$, then $\frac{d}{dx}\arcsin x = \frac{1}{\cos(\arcsin x)}$

what is $\cos(\arcsin x)$?

angle of a right triangle that has $\frac{\text{opp}}{\text{hyp}} = x$ (differentiating inverse trig functions)

$x^2 + \text{adj}^2 = 1^2$

$\Rightarrow \text{adj}^2 = 1-x^2$

$\Rightarrow \text{adj} = \sqrt{1-x^2}$

$\therefore \cos(\arcsin x) = \sqrt{1-x^2}$

14) $\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$

15) $\frac{d}{dx}\arccos x = \frac{-1}{\sqrt{1-x^2}}$

16) $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$

Lec 9 - Linearization & related rates

Linear approximation: The tangent line of f(x) at "a" is called the linear approximation (or linearization, or tangent approximation to f(x) centred at x=a, and it is $L(x) = f'(a)(x-a) + f(a)$

Note: We call this an approximation of f(x) because it's the line that most closely ressembles f(x) at x=a: they have $L(a) = f(a)$ and $L'(a) = f'(a)$ which is all we could ask of a line.

Ex: What's the equation of the tangent line to f(x) at x=a?

we always take the same steps. Sub $m = f'(a)$, $x_0 = a$, $y_0 = f(a)$

formula: $y = m(x-x_0) + y_0 \rightarrow y = f'(a)(x-a) + f(a)$

Ex: linear approximation to estimate $\sin(0.1)$.

It may be tempting to choose a=0.1, but we should choose "a" such that $\sin(a)$ is easy to compute by hand.

Let's choose a=0, which is pretty close to 0.1.

Let $f(x) = \sin x$, $a = 0$

$f'(x) = \cos x$ $L(x) = f'(a)(x-a) + f(a)$

$f(0) = \sin(0) = 0$ $= 1(x-0) + 0$

$f'(0) = \cos(0) = 1$ $= x$

so according to this, $L(x) = x$ will pretend to be $f(x) = \sin x$ as long as x is near 0 (the a-value)

to estimate $\sin(0.1)$, we just use $L(0.1) \approx 0.1$

Easy! Compare to the actual answer: $\sin(0.1) \approx 0.0998$

Ex: $f(x) = \sqrt{x+3}$. f(x) at a=1 to estimate $\sqrt{3.98}$ & $\sqrt{4.05}$

$\lim_{n\to 0}\frac{\sin n}{n} = 1$ we use a=1 because $f(1) = \sqrt{3}$ is easy to compute, and $f(a)$ is close to $\sqrt{3.98}$ and $\sqrt{4.05}$

$\sin x \approx x$ when $x \approx 0$

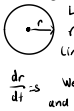
$f'(x) = \frac{1}{2\sqrt{x+3}}$, so $f'(1) = \frac{1}{2\sqrt{1+3}} = \frac{1}{4}$ and $f(1) = \sqrt{4} = 2$

$\therefore L(x) = f'(a)(x-a) + f(a) = \frac{1}{4}(x-1) + 2$

then $\sqrt{3.98} = f(0.98) \approx L(0.98) = \frac{1}{4}(0.98-1) + 2 = -0.005 + 2 = 1.995$

and $\sqrt{4.05} = f(1.05) \approx L(1.05) = \frac{1}{4}(1.05-1) + 2 = 0.0125 + 2 = 2.0125$

Ex: The radius of a circle is increasing at a rate of 5cm/s. At what rate is the area increasing when the radius is 20 cm?



Let A be the area of the circle (in cm^2)

r be the radius (in cm), and t be time (in s).

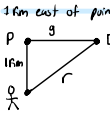
$\frac{dr}{dt} = 5$ We know the relationship between A and r is $A = \pi r^2$, but we want to know $\frac{dA}{dt}$, the rate of change of the area with respect to time we can differentiate $A = \pi r^2$ w.r.t. t to write $\frac{dA}{dt} = 2\pi r\cdot\frac{dr}{dt}$. As with implicit differentiation, each time we need the derivative of A or r, we write $\frac{dA}{dt}$ or $\frac{dr}{dt}$, respectively. We were given $\frac{dr}{dt} = 5$

If we want to sub r=20, we could write

$\left.\frac{dA}{dt}\right|_{r=20} = 2\pi\cdot 20\cdot\left.\frac{dr}{dt}\right|_{r=20} = 2\pi\cdot 20\cdot 5 = 200\pi\ cm^2/s$

the area is increasing by $200\pi\ cm^2/s$ at that moment

Ex: You are standing 1 km south of point P, and in the distance you can see Howard Bousseman driving his car 75 km/h headed east, away from point P. How quickly is the distance between you and Howard Bousseman changing at the moment he is 1 km east of point P?



Let r be the distance between us and H.B, and let g be the distance between point P and H.B.

We're trying to find $\frac{dr}{dt}$

we can write $r^2 = 1^2 + g^2$ (pythagoras)

$\Rightarrow r = \sqrt{1+g^2}$

Differentiating w.r.t. t gives

$$\frac{dr}{dt} = \frac{1}{2\sqrt{1+g^2}}\cdot 2g\cdot\frac{dg}{dt}$$

$$\left.\frac{dr}{dt}\right|_{g=1} = \frac{1}{2\sqrt{1+1^2}}\cdot 2\cdot 1\cdot 75 = \frac{75}{\sqrt{2}}$$

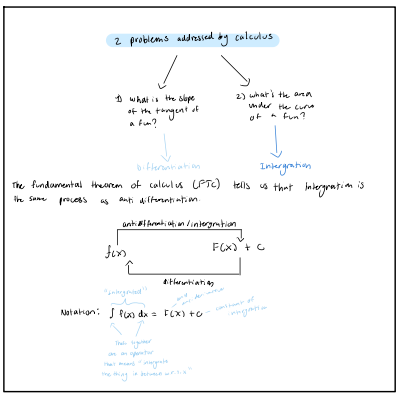
$\therefore$ The distance between us is increasing by $\frac{75}{\sqrt{2}}$ km/h

Lec 10 – Antiderivatives & Areas under curves

Antiderivative: A function F is called an antiderivative of f if $F'(x) = f(x)$ that is, $\frac{d}{dx} F(x) = f(x)$

(antiderivative of $f(x)$ ↘ derivative of $F(x)$)

Note: we can always add a constant to get a new antiderivative, but that's it. therefore, we are often trying to find the family of antiderivatives (or the general antiderivative) of a fun, acknowledging that any constant could be added on. Example: the general antiderivative of $3x^2$ is $x^3 + C$ where C is understood to be any/every constant.



Basic rules of integration

1. for any constant $K \in \mathbb{R}$: $\int K\,dx = Kx + C$
2. For any power $n \in \mathbb{R}$ except for $n = -1$ $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (opposite of power rule)
3. $\int x^{-1} dx = \ln|x| + C$ ($\ln|x| = \begin{cases} \ln x; & x > 0 \\ \ln(-x); & x < 0 \end{cases}$) $\frac{d}{dx} \ln|x| = \frac{1}{x}$?
4. $\int K f(x)\,dx = K \int f(x)\,dx$
5. $\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$
6. $\int \cos x\,dx = \sin x + C$
7. $\int \sin x\,dx = -\cos x + C$
8. $\int \sec^2 x\,dx = \tan x + C$
9. $\int e^x dx = e^x + C$
10. $\int a^x dx = \frac{1}{\ln a} \cdot a^x + C$
11. $\int \frac{1}{1+x^2} dx = \arctan x + C$
12. $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$
13. $\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan(x/a) + C$

Definition: (Left-Endpoint, Right-Endpoint, Midpoint, Sample Point Rules)

Let f be a continuous function on $[a,b]$. Suppose we want to estimate the area under $y = f(x)$ with n approximating rectangles between $x=a$ and $x=b$. let $\Delta x = \frac{b-a}{n}$.

left-endpoint approximation

$$L_n = \Delta x [f(a) + f(a+\Delta x) + f(a+2\Delta x) + \dots + f(a+(n-1)\Delta x)]$$

($k=1$, $k=2$, $k=3$, $k=n$)

$$= \Delta x \sum_{k=1}^{n} f(a+(k-1)\Delta x)$$

sigma notation: "sum form $k=1$ to n of ..."

Add up the terms that have the form $f(a+(k-1)\Delta x)$, first with $k=1$, then $k=2$ and so on until the last term which has $k=n$.

right-endpoint approximation

$$R_n = \Delta x \sum_{k=1}^{n} f(a+k\Delta x)$$

midpoint approximation

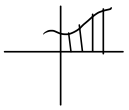
$$M_n = \Delta x \sum_{k=1}^{n} f(a+(k-\tfrac{1}{2})\Delta x)$$

halfway in between left-and right-endpoints

Lec 11 – Definite integrals & the fundamental theorem of calculus

Riemann Sum

Let f be a cts on $[a,b]$. A Riemann Sum is any sum $S_n = \sum_{k=1}^{n} \Delta x_k f(x_k^*)$, where $[a,b]$ is subdivided into sub intervals $[x_0,x_1], [x_1,x_2], \dots, [x_{n-1},x_n]$ with $x_0 = a$ and $x_n = b$; $\Delta x_k = x_k - x_{k-1}$ and x_k^* is any "sample point" on $[x_{k-1}, x_k]$.



[Note: in this course we'll only ever consider rectangles where the Δx_k's are equal.] the the area under f in $[a,b]$ is $\lim_{n\to\infty} S_n$

Definite Integrals

Let f be either cts on $[a,b]$ or cts on $[a,b]$ except for finitely many jump discontinuities.

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} S_n$$

(upper limit of integration; lower limit) where S_n is a Riemann Sum on $[a,b]$.

If this limit exists, then f is integrable on $[a,b]$.

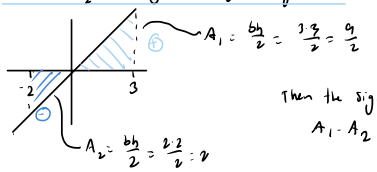
(Note: $\int f(x)\,dx$ is called a indefinite integral)

Signed area

The definite integral computes the area "under" the curve $y=f(x)$. In regions where $f(x)$ is negative, the area computation with negative, too. This computation is different from the usual notion of area so we call it signed area or net area (usual area $= \int_a^b |f(x)|\,dx$)

ex:

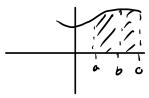
evaluate $\int_{-2}^{3} x\,dx$ by computing the signed area



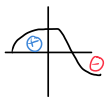
Then the signed area is $A_1 - A_2 = \frac{9}{2} - 2 = \frac{5}{2}$

Rules of definite integrals

1. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
2. $\int_a^a f(x)\,dx = 0$
3. $\int_a^b K f(x)\,dx = K \int_a^b f(x)\,dx$
4. $\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$
5. $\int_a^b f(x)\,dx + \int_d^c f(x)\,dx = \int_a^c f(x)\,dx$
6. If $f(x) \le g(x)$ on $[a,b]$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$.

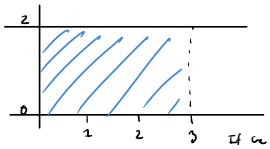


Recall: definite integral computes signed area. It's clear that the area is just a number



But we could come up with another fun, $F(x)$, that tells us the area under $y=f(x)$ between e.g. 0 and x.

Example: $f(x) = 2$ the area under f between 0 and any x-value is just 2 times that x value. we can write this like $F(x) = \int_0^x f(t)\,dt = 2x$



If we choose $x=3$, the area is $2 \cdot 3$.

(F: our area fun; f: area between 0 and x; we use these not x, as a "dummy" variable. we need to keep it seperate from x because x is what tells F how much area to compute.)

Note that $F(x) = \frac{d}{dx} 2x = 2 = f(x)$

Is this a coincidence? NO

Theorem: Suppose f is cts on $[a,b]$. Then $F(x) = \int_a^x f(t)\,dt$ is cts on $[a,b]$ and diff on (a,b) and we have $F'(x) = \frac{d}{dx} \int_a^x f(t)\,dt = f(x)$.

Remark.

Note: This is just a differentiation rule in which the lower limit of integration must be a constant and the upper limit must be x.

ex: $\frac{d}{dx} \int_0^{x^2} f(t)\,dt$

FTC I + chain rule say that

$\frac{d}{dx} \int_0^{x^2} f(t)\,dt = f(x^2) \cdot \frac{d}{dx} x^2 = f(x^2) \cdot 2x$

(the FTC I rule with x^2 subbed in; double chain rule)

Remember, FTC I needs "x" in the upper limit, but there was x^2 there instead. That's why this required chain rule.

ex: $\frac{d}{dx} \int_x^{x^2} f(t)\,dt$ $\int_x^0 f(t)\,dt + \int_0^{x^2} f(t)\,dt = \int_x^{x^2} f(t)\,dt$

we can't apply FTC I yet because we need a constant in the lower limit. But we can rewrite

$\frac{d}{dx} \left(\int_x^0 f(t)\,dt + \int_0^{x^2} f(t)\,dt \right)$

$= \frac{d}{dx} \left(-\int_0^x f(t)\,dt + \int_0^{x^2} f(t)\,dt \right)$

$= \underbrace{-f(x)}_{\text{follows FTC I exactly}} + \underbrace{f(x^2) \cdot 2x}_{\text{see (c)}}$

Lec 12 – Substitution

Ex:

a) $\int 2e^{2x}\,dx = e^{2x} + C$ b) $\int 2e^{2x+1}\,dx = e^{2x+1} + C$ c) $\int 2x e^{x^2+1}\,dx = e^{x^2+1} + C$

d) $\int 3(x+1)^2\,dx = (x+1)^3 + C$ e) $\int (x+1)^3\,dx = \frac{(x+1)^4}{4} + C$ f) $\int 3x^2 (x^3+1)^5\,dx = \frac{(x^3+1)^6}{6} + C$

g) $\int \frac{g'(x)}{g(x)}\,dx = \ln|g(x)| + C$

chain rule

↳ $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) \rightarrow f(g(x)) = \int f'(g(x)) g'(x)\,dx$

lesson

when faced with an integrand that we suspect is the result of a chain rule, we must use substitution.

Step 1) let $u = g(x)$ for an appropriate choice of $g(x)$ (it may take multiple tries to get it right)

Step 2) Find $\frac{du}{dx} = g'(x)$

Step 3) "multiply by dx" and isolate it: $\frac{du}{g'(x)} = dx$

Step 4) substitute $dx = \frac{du}{g'(x)}$ (except in rare cases).

Step 5) substitute $g(x) = u$ into the integrand as needed

pray $\int f'(g(x)) g'(x)\,dx = \int f'(u) g'(x) \cdot \frac{du}{g'(x)} = \int f(u)\,du$. If we get this then hopefully we can obtain $f(u) + C$.

Step 6) Back substitute $u = g(x)$ for your final answer

Ex: $\int 2x e^{x^2+1}\,dx = \int 2x e^{x^2+1} \frac{du}{dx}$

sub

$= \int e^u\,du$

Ex: $\int x^2 e^{x^3-9}\,dx = \int x^2 e^{x^3-9} \frac{du}{3x^2}$

sub

$= \frac{1}{3} \int e^u\,du$

Ex: $\int \frac{(\sqrt{x}-\sqrt{2})^8}{\sqrt{x}}\,dx = \int \frac{u^8}{\sqrt{x}} (-2\sqrt{x}\,du)$

sub

$= -2 \int u^8\,du$

$= -2 \cdot \frac{u^9}{9} + C$

Let $u = x^2+1$
$\Rightarrow \frac{du}{dx} = 2x$
$\Rightarrow dx = \frac{du}{2x}$

$= e^u + C$
$= e^{x^2+1} + C$

Let $u = x^3 - 9$
$\Rightarrow \frac{du}{dx} = 3x^2$
$\Rightarrow dx = \frac{du}{3x^2}$

$= \frac{1}{3} e^u + C$
$= \frac{1}{3} e^{x^3-9} + C$

Let $u = 3 - \sqrt{x}$
$\Rightarrow \frac{du}{dx} = \frac{-1}{2\sqrt{x}}$
$\Rightarrow dx = -2\sqrt{x}\, du$

$= -2 \int \sqrt{u}\,(3-u)\, du$
$= -2 \int (3u^{1/2} - u^{3/2})\, du$
$= -2 (2u^{3/2} - \frac{2}{5}u^{5/2}) + C$
$= -2(3-\sqrt{x})^{3/2} + C$

Ex: $\int \frac{x}{\sqrt{x}+2}\, dx$

Sub
Let $u = \sqrt{x} + 2$
$\Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}}$
$\Rightarrow dx = 2\sqrt{x}\, du$

$= \int \frac{x}{u} \cdot 2\sqrt{x}\, du$
$= 2 \int \frac{(u-2)^2}{u} (u-2)\, du$
$= 2 \int \frac{(u-2)^3}{u}\, du$
$= 2 \int \frac{u^3 - 2 \cdot 3u^2 + 4 \cdot 3u - 8}{u}\, du$
$= 2 \int (u^2 - 6u + 12 - \frac{8}{u})\, du$
$= 2 (\frac{u^3}{3} - 3u^2 + 12u - 8 \ln|u|) + C$

we also have $u - 2 = \sqrt{x}$
and $(u-2)^2 = x$

+ there is more

Sub
Let $u = \sqrt{x}$
$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$
$\Rightarrow dx = 2\sqrt{x}\, du$

$= \int \frac{x}{u+2} \cdot 2\sqrt{x}\, du$

we also have $u^2 = x$
$= 2 \int \frac{u^2}{u+2}\, du$

Ex: $\int \cos^2 x \sin^3 x\, dx$

we can convert cosx to sinx or vice-versa two powers at a time by using $\cos^2 x = 1 - \sin^2 x$ or $\sin^2 x = 1 - \cos^2 x$.

Let's use $\sin^2 x = 1 - \cos^2 x$ one time to get

$I = \int \cos^2 x (1 - \cos^2 x) \sin x\, dx$

Sub
Let $u = \cos x$
$\frac{du}{dx} = -\sin x$
$\Rightarrow dx = \frac{-du}{\sin x}$

$= \int u^2 (1-u^2) \sin x \cdot \frac{-du}{\sin x}$
$= -\int (u^2 - u^4)\, du$
$= -(\frac{u^3}{3} - \frac{u^5}{5}) + C$
$= \frac{-\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$

In general, to solve $\int \cos^m x \sin^n x\, dx$:

- If m is odd:
  reserve one copy of cosx and turn rest into sinx by using $\cos^2 x = 1 - \sin^2 x$.

Ex: $\int \sin^4 x \cos^2 x\, dx$

We must use the half-angle identities:

$I = \int [\frac{1}{2}(1 - \cos(2x))]^2 \cdot \frac{1}{2}(1 + \cos(2x))\, dx$
$= \frac{1}{8} \int (1 - 2\cos(2x) + \cos^2(2x))(1 + \cos(2x))\, dx$
$= \cdots$
$= \frac{1}{8} [\,(1 - \cos(2x) - \cos^2(2x) + \cos^3(2x))\, dx$

normal; use half-angle again; add on of $\cos(2x)$

theorem (FTC2): If f is cts on $[a,b]$ and F is any antiderivative of f on $[a,b]$, $\int_a^b f(x)\,dx = F(b) - F(a)$

Note: FTC2 gives us a concrete way to solve definite integrals.

ex:
use FTC2 to evaluate $\frac{d}{dx} \int_a^{x^2} f(t)\, dt$, where $f(t) = 2$.

$F(x) = 2x$

$\frac{d}{dx} \int_x^{x^2} f(t)\, dt = \frac{d}{dx}(F(x^2) - F(x)) = \frac{d}{dx}[F(t)]_x^{x^2}$

$= f(x^2) \cdot 2x - f'(x)$

Lec 13 - Integration by parts

Recall the product rule for differentiation:

$\frac{d}{dx} f(x)g(x) = f'(x)g(x) + f(x)g'(x)$

Then integrating both sides gives

$f(x)g(x) = \int f'(x)g(x)\, dx + \int f(x)g'(x)\, dx$

$\Rightarrow \int f(x)g'(x)\, dx = f(x)g(x) - \int f'(x)g(x)\, dx$ (rearranged)

this doesn't necessarily solve our problem, but it lets us turn the integral on the left into a (hopefully) easier one on the right

A guide to integration by parts:

Given a problem of the form:
$\int f(x)g'(x)\, dx = ?$
choose variable u and v such that:
$u = f(x)$
$dx = g'(x)\, dx$
now the original expression becomes
$\int u\, dv = ?$
which definitely looks easier

Step 1) split the integrated into two factors, $u = f(x)$ and $dv = g'(x)\, dx$

Step 2) Differentiate u to get $\frac{du}{dx} = f'(x)$
$\Rightarrow du = f'(x)\, dx$. Anti-differentiate dv to get $v = g(x)$.

Step 3) write $\int u\, dv = uv - \int v\, du$, immediately replacing u, v, du, dv with their respective funs in x.

Ex: $\int x e^x\, dx$

$u = x \quad v = e^x$
$du = dx \quad dv = e^x dx$

IBP

$\int u\, dv = uv - \int v\, du$
$\int x e^x\, dx = xe^x - \int e^x\, dx$
$= xe^x - e^x + C$

check by differentiating:
$\frac{d}{dx}(xe^x - e^x + C) = e^x + xe^x - e^x = xe^x$
things would go wrong if we tried something else:

IBP
$u = e^x \quad v = \frac{x^2}{2}$
$du = e^x dx \quad dv = x\, dx$
$\int x e^x\, dx = \frac{x^2}{2} e^x - \int \frac{x^2}{2} e^x\, dx$
worse than before

We can use a priority list to determine what should go into u.

I - inverse trig (arcsinx, arctanx)
L - logarithms
A - algebra (powers of x)
T - trig (sinx, cosx, etc.)
E - exponential (e^x, a^x)

} definitely put these into u!
} rarely do we want to put these into u.

Ex: $\int \ln x\, dx$

IBP
$u = \ln x \quad v = x$
$du = \frac{1}{x} dx \quad dv = dx$

$= x \ln x - \int \frac{x}{x}\, dx$
$= x \ln x - \int 1\, dx$
$= x \ln x - x + C$

1) $\int x^n e^x\, dx = x^n e^x - \int n x^{n-1} e^x\, dx$ (reduces power by 1)

IBP
$u = x^n \quad v = e^x$
$du = nx^{n-1} dx \quad dv = e^x dx$

2) $\int x^n \sin x\, dx = -x^n \cos x + \int n x^{n-1} \cos x\, dx$ (reduces power of x by 1)

$u = x^n \quad v = -\cos x$
$du = nx^{n-1} dx \quad dv = \sin x\, dx$

3) $\int x^n (\ln x)^m\, dx$

$u = (\ln x)^m \quad v = \frac{x^{n+1}}{n+1}$
$du = m(\ln x)^{m-1} \frac{1}{x} dx \quad dv = x^n dx$

IBP

$= \frac{x^{n+1}(\ln x)^m}{n+1} - \int \frac{m}{n+1} \cdot x^n (\ln x)^{m-1} dx$

(reduces power of $\ln x$ by 1).

Ex: $\int \arctan(\sqrt{x})\, dx$

Sub
$u = \sqrt{x}$
$\Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}}$
$\Rightarrow dx = 2\sqrt{x}\, du$

$= \int 2\sqrt{x} \arctan(\sqrt{x})\, du$
$= 2 \int u \arctan(u)\, du$
$= 2 [\frac{u^2}{2} \arctan(u) - \frac{1}{2} \int \frac{u^2}{1+u^2}\, du]$
$= u^2 \arctan(u) - \int \frac{1+u^2-1}{1+u^2}\, du$
$= u^2 \arctan(u) - \int \frac{1+u^2}{1+u^2}\, du + \int \frac{1}{1+u^2}\, du$
$= u^2 \arctan(u) - u + \arctan(u) + C$
$= x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x}) + C$

IBP
$u = \arctan(u) \quad v = \frac{u^2}{2}$
$du = \frac{1}{1+u^2} du \quad dv = u\, du$

Lec 14 - Trigonometric substitution

Ex 14.1

$\int_{-1}^{1} \frac{1}{4x^2+1}\, dx$

[Structure: $\frac{1}{a^2x^2+1}$ Idea: try to get $\int \frac{1}{u^2+1}\, du$]

Sub
$u = 2x$
$\frac{du}{dx} = 2$
$\Rightarrow dx = \frac{1}{2} du$

$I = \int_{x=-1}^{1} \frac{1}{u^2+1} \cdot \frac{1}{2}\, du$

when we substitute into a definite integral, we can leave the bounds of integration in terms of x, or we can rewrite them in terms of u. By default (if no variable is mentioned), they are in terms of the subject of integration. Let's change them to u:

when $x = -1$: $u = 2x = -2$
when $x = 1$: $u = 2x = 2$
when $I = \int_{-2}^{2} \frac{1}{u^2+1}\, du$
$= \frac{1}{2} [\arctan u]_{-2}^{2}$
$= \frac{1}{2} (\arctan(2) - \arctan(-2))$
(this doesn't simplify well, so we can leave it.)

Ex 14.2

[Structure: $\frac{1}{x^2+a^2}$ Idea: factor out a^2 from denom. To get $\frac{1}{a^2} \cdot \frac{1}{(\frac{x}{a})^2+1}$, then use previous technique]

Sub
$u = \frac{x}{2}$
$\frac{du}{dx} = \frac{1}{2}$
$\Rightarrow dx = 2\, du$

$I = \frac{1}{4} \int \frac{1}{\frac{1}{4}x^2+1}\, dx$
$= \frac{1}{4} \int \frac{1}{(\frac{x}{2})^2+1}\, dx$
$= \frac{1}{4} \int \frac{1}{u^2+1} \cdot 2\, du$
$= \frac{1}{2} \int \frac{1}{u^2+1}\, du$
$= \frac{1}{2} \arctan(u) + C$
$= \frac{1}{2} \arctan(\frac{x}{2}) + C$

Ex 14.3

$\int \frac{1}{x^2+6x+8}\, dx$

[Structure: $\frac{1}{ax^2+bx+c}$ Idea: complete the square to get $\frac{1}{u^2+1}$ and ax^2+bx+c cannot be factored (it is irreducible)]

Recall completing the square: $(x+3)^2 = x^2+6x+9$, which matches coefficients of x^2 and x. Then add 4 afterward to recover what we had.

$I = \int \frac{1}{(x+3)^2+8-9}\, dx$
$= \frac{1}{4} \int \frac{1}{(\frac{x+3}{2})^2+1}\, dx$
$= \frac{1}{4} \int \frac{1}{u^2+1} \cdot 2\, du$
$= \frac{1}{2} \int \frac{1}{u^2+1}\, du$
$= \frac{1}{2} \arctan(u) + C$

Sub
$u = \frac{x+3}{2}$
$\frac{du}{dx} = \frac{1}{2}$

Ex 14.4

$\int \frac{4x+1}{x^2+2x+4}\, dx$

[Structure: $\frac{Ax+B}{ax^2+bx+c}$ Idea: split into two fractions and complete the square, where ax^2+bx+c is irreducible.]

Note that if u = denominator $= x^2+2x+4$
Then $du = (2x+2)\, dx$

we want to split up the numerator so we can use this substitution on it. Make numerator a multiple of $2x+2$ plus whatever constant is left over.

$I = \int \frac{4x+4-3}{x^2+2x+4}\, dx = \int \frac{4x+4}{x^2+2x+4}\, dx - 3 \int \frac{1}{x^2+2x+4}\, dx$

using sub. ↓ complete the square ↓

Sub
$v = \frac{x+1}{\sqrt{3}}$

$= \int \frac{4x+4}{u} \cdot \frac{du}{2x+2} - 3 \int \frac{1}{(x+1)^2+3}\, dx$
$= 2 \int \frac{1}{u}\, du - 3/3 \int \frac{1}{(\frac{x+1}{\sqrt{3}})^2+1}\, dx$

$= \frac{1}{2}\arctan\left(\frac{x+2}{2}\right) + c$ $\Rightarrow dx = 2\,du$

Example 5: $\int_0^{3/2} \frac{x^2}{\sqrt{4-x^2}}\,dx$

[structure: $\sqrt{a^2-x^2}$ in denom; idea: try to use identity $1-\sin^2\theta = \cos^2\theta$]

In general, sub $x = a\sin\theta$, $-\pi/2 \le \theta \le \pi/2$

Trig sub: $x = 2\sin\theta$, $-\pi/2 \le \theta \le \pi/2$

So $x^2 = 4\sin^2\theta$

$\Rightarrow dx = 2\cos\theta\,d\theta$

Note, x is written in terms of the new variable!

$\theta = \arcsin\left(\frac{x}{2}\right)$

when $x = 3/2$: $\theta = \arcsin(1/2)$

when $x = 0$: $\theta = \arcsin(0)$

$$I = \int_0^{\pi/6} \frac{x^2}{\sqrt{4-x^2}}\,2\cos\theta\,d\theta$$
$$= \int_0^{\pi/6} \frac{4\sin^2\theta \cdot 2\cos\theta}{\sqrt{4-4\sin^2\theta}}\,d\theta$$
$$= \frac{8}{2}\int_0^{\pi/6} \frac{\sin^2\theta\cos\theta}{\sqrt{1-\sin^2\theta}}\,d\theta$$
$$= 4\int_0^{\pi/6} \frac{\sin^2\theta\cos\theta}{\sqrt{\cos^2\theta}}\,d\theta \quad (\sqrt{\cos^2\theta} = |\cos\theta| = \cos\theta \text{ when } -\pi/2 \le \theta \le \pi/2)$$
$$= 4\int_0^{\pi/6} \sin^2\theta\,d\theta$$
$$= 4\cdot\tfrac{1}{2}\int_0^{\pi/6} (1-\cos(2\theta))\,d\theta$$
$$= 2\left[\left(\tfrac{\pi}{6} - \tfrac{1}{2}\sin\left(\tfrac{\pi}{3}\right)\right) - \left(0 - \tfrac{1}{2}\sin(0)\right)\right]$$
$$= 4\left(\tfrac{\pi}{6} - \tfrac{1}{2}\cdot\tfrac{\sqrt{3}}{2}\right) = \tfrac{\pi}{3} - \tfrac{\sqrt{3}}{4}$$

Ex 14.6

$\int \frac{1}{\sqrt{4+x^2}}\,dx$

[structure: $\sqrt{a^2+x^2}$ in denom; idea: use identity $1+\tan^2\theta = \sec^2\theta$]

In general: sub $x = a\tan\theta$

Trig sub

$x = 2\tan\theta$, $-\pi/2 < \theta < \pi/2$

So $x^2 = 4\tan^2\theta$

$\Rightarrow dx = 2\sec^2\theta\,d\theta$

$$I = \int \frac{1}{\sqrt{4+4\tan^2\theta}} \cdot 2\sec^2\theta\,d\theta$$
$$= \tfrac{2}{2}\int \frac{\sec^2\theta}{\sqrt{1+\tan^2\theta}}\,d\theta$$
$$= \int \frac{\sec^2\theta}{\sqrt{\sec^2\theta}}\,d\theta$$
$$= \int \sec\theta\,d\theta$$

to be continued...

Ex 14.7.

$$\int \frac{1}{\sqrt{25x^2-4}}\,dx$$

[structure: $\sqrt{x^2-a^2}$ in denom; Idea: try to use identity $\sec^2\theta - 1 = \tan^2\theta$]

You'll need to deal with the coeff. in front of x?

In general, to solve integrals of the form $\int \tan^m x \sec^n x\,dx$:

recall our differentiation rules:

$\left[\frac{d}{dx}\tan x = \sec^2 x \quad \frac{d}{dx}\sec x = \sec x \tan x\right]$

- If n is even and $n \ge 4$:
  Reserve one copy of $\sec^2 x$ and convert the rest to $\tan x$ using $\sec^2 x = \tan^2 x + 1$.
  Then Sub $u = \tan x$
- If m is odd and $m, n \ge 1$
  Reserve one copy of $\sec x \tan x$ and convert all the $\tan x$ to $\sec x$ using $\tan^2 x = \sec^2 x - 1$. Then sub $u = \sec x$

Ex: $\int \tan^3 x \sec^4 x\,dx$ ($m=3$, $n=4$)

Either strategy will work, so let's use the first one

$I = \int \tan^3 x \sec^2 x (\tan^2 x + 1)\,dx$

Sub $u = \tan x$, $\frac{du}{dx} = \sec^2 x$, $\Rightarrow dx = \frac{du}{\sec^2 x}$

$$= \int u^3 \sec^2 x (u^2+1) \cdot \frac{du}{\sec^2 x}$$
$$= \int (u^5 + u^3)\,du$$
$$= \frac{u^6}{6} + \frac{u^4}{4} + C$$
$$= \frac{\tan^6 x}{6} + \frac{\tan^4 x}{4} + C$$

Lec 15 - partial fractions

Ex 15.1

Integrate $I = \int \frac{x+5}{x^2+x-2}\,dx$

Nothing we learned seems to work.

But observe that $\frac{x+5}{x^2+x-2} = \frac{x+5}{(x+2)(x-1)}$

Partial fractions will always allow us to write this as

$\frac{A}{x+2} + \frac{B}{x-1}$ for some $A, B \in \mathbb{R}$

because if we cross multiply, we should be able to get the same thing we started with.

$$\frac{A}{x+2} + \frac{B}{x-1} = \frac{A(x-1) + B(x+2)}{(x+2)(x-1)}$$
$$= \frac{Ax - A + Bx + 2B}{(x+2)(x-1)}$$
$$= \frac{(A+B)x + (-A+2B)}{(x+2)(x-1)}$$
$$= \frac{x+5}{(x+2)(x-1)}$$

By matching coefficients, we see that

$$\left.\begin{aligned} A + B &= 1 \\ -A + 2B &= 5 \end{aligned}\right\} A = 1 - B \left.\begin{aligned} \\ -(1-B) + 2B &= 5 \end{aligned}\right\} \begin{aligned} 3B &= 6 \Rightarrow B = 2 \\ \text{Then } A &= 1 - 2 = -1 \end{aligned}$$

2) $\frac{3}{7}B_1$

$\frac{7}{2} = 3 + \frac{1}{2}$

so our integral becomes

$I = \int \frac{-1}{x+2}\,dx + \int \frac{2}{x-1}\,dx = -\ln|x+2| + 2\ln|x-1| + C$

Partial fraction decomposition

When our integrand is a rational function $\frac{P(x)}{Q(x)}$ where P & Q are polynomials, we can use this technique.

Step 1: If the degree of $P \ge$ the degree of Q, then we must use long division first to get $P_1(x) + \frac{P_2(x)}{Q(x)}$ ← remainder after long division

Step 2: Now that the degree of P (or P_2) < the degree of Q, we can fully factor $Q(x) = Q_1(x)^{r_1} \cdot Q_2(x)^{r_2} \cdot \ldots \cdot Q_n(x)^{r_n}$

where each $Q_i(x)$ is either a linear factor or an irreducible polynomial of degree at least 2 (for our purposes, it will always be an irreducible quadratic).

Note: each r_i is called the multiplicity of $Q_i(x)$, the number of times that factor appears in the factorization.

Now there are four possibilities for each factor $Q_i(x)^{r_i}$:

Case (a): $Q_i(x) = ax+b$ is linear and $r_i = 1$.

Then this produces a term $\frac{A}{ax+b}$ in the decomposition e.g. if our fun is $\frac{x+3}{\underbrace{(x+2)}_{Q_1(x)}\underbrace{(x-5)}_{Q_2(x)}}$

Then we get $\frac{A}{x+2} + \frac{B}{x-1}$

Case (b): $Q_i(x) = ax+b$ is linear and $r_i > 1$.

Then r_i terms like $\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \ldots + \frac{A_{r_i}}{(ax+b)^{r_i}}$

e.g. if our fun is $\frac{x+5}{\underbrace{(x+2)^2}_{[Q_1(x)]^{r_i}}\underbrace{(x-1)}_{Q_2(x)}}$

Then we get $\frac{A_1}{x+2} + \frac{A_2}{(x+2)^2} + \frac{B}{x-1}$

Case (c): $Q_i(x) = ax^2+bx+c$ is an irreducible quadratic and $r_i = 1$.

Then a term like $\frac{Ax+B}{ax^2+bx+c}$ appears.

e.g. $f(x) = \frac{x+5}{\underbrace{(x^2+x+1)}_{Q_1(x)}\underbrace{(x-1)}_{Q_2(x)}}$

then we get $\frac{Ax+B}{x^2+x+1} + \frac{C}{x-1}$

Case (d): $Q_i(x) = ax^2+bx+c$ is an irreducible quadratic and $r_i > 1$.

Then r_i terms like $\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \ldots + \frac{A_{r_i}x+B_{r_i}}{(ax^2+bx+c)^{r_i}}$

appear.

e.g. $f(x) = \frac{x+5}{\underbrace{(x^2+x+1)^2}_{[Q_1(x)]^{r_i}}\underbrace{(x-1)}_{Q_2(x)}}$

we get $\frac{A_1x+B_1}{x^2+x+1} + \frac{A_2x+B_2}{(x^2+x+1)^2} + \frac{C}{x-1}$

Ex 15.2

$$\frac{1}{x^2-9} = \frac{1}{(x+3)(x-3)}$$

$$\frac{A}{x+3} + \frac{B}{x-3}$$

$$\frac{1}{(x+3)(x-3)} = \frac{A}{x+3} + \frac{B}{x-3}$$

$$1 = A(x-3) + B(x+3)$$
$$= Ax - 3A + Bx + 3B$$
$$= x(A+B) + 3(B-A)$$

$A+B=0 \Rightarrow A=-B$

$B-A=\frac{1}{3} \quad A=-\frac{1}{6}$

$B-(-B)=\frac{1}{3}$

$2B=\frac{1}{3}$

$B=\frac{1}{6}$

$$\frac{-1}{6(x+3)} + \frac{1}{6(x-3)}$$

$$-\frac{1}{6}\int \frac{1}{x+3}\,dx + \frac{1}{6}\int \frac{1}{x-3}\,dx$$

$$-\frac{1}{6}\cdot \ln|x+3| + \frac{1}{6}\cdot \ln|x-3|$$

Ex 15.3

$$\frac{x^4-2x^2+4x+1}{x^3-x^2-x+1} = x+1 + \frac{4x}{x^3-x^2-x+1}$$

$(x-1)^2(x+1)$

$$\begin{array}{r|l} x^4-2x^2+4x+1 & x^3-x^2-x+1 \\ -(x^4-x^3-x^2+x) & \hline x+1 \\ \hline x^3-x^2+3x+1 & \\ -(x^3-x^2-x+1) & \\ \hline 4x & \end{array}$$

$$\frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{x+1} = \frac{4x}{x^3-x^2-x+1}$$

$$x+1 + \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+1)}$$
$$x+1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{(x+1)}$$

$$A(x-1)(x+1) + B(x+1) + C(x-1)^2 = 4x$$
$$A(x^2-1) + B(x+1) + C(x^2-2x+1) = 4x$$
$$Ax^2 - A + Bx + B + Cx^2 - 2Cx + C = 4x$$
$$x^2(A+C) + x(B-2C) + (B+C-A) = 4x$$

$A+C=0 \rightarrow (B+C)+C=0 \quad B+2C=0 \Rightarrow B=-2C$

$B-2C=4 \rightarrow -B+B=4 \rightarrow 2B=4 \rightarrow B=2$

$B+C-A=0 \Rightarrow B+C=A$

$2+C=A$

$2C=-2$

$C=-1$

$2+C+C=0$

$2+2C=0$

$2-1-A=0$

$1-A=0$

$1=A$

$A=1$

$B=2$

$C=-1$

$$\int x\,dx + \int 1\,dx + \int \frac{1}{x-1}\,dx + 2\int \frac{1}{(x-1)^2}\,dx - \int \frac{1}{x+1}\,dx$$

$$\frac{x^2}{2} + x + \ln|x-1| + \frac{2}{3}(x-1)^{-3} - \ln|x+1|$$

Ex 15.4

$$\int \frac{2x^2-x+4}{x^3+4x}\,dx$$

$$\frac{2x^2-x+4}{x(x^2+4)}$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+C$$

$$\frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$(A)(x^2+4) + (Bx+C)(x) = 2x^2-x+1$$
$$Ax^2 + 4A + Bx^2 + Cx = 2x^2-x+1$$
$$x^2(A+B) + x(C) + 4(A) = 2x^2-x+1$$

$A+B=2 \rightarrow \frac{1}{4}+B=2 \quad B=\frac{7}{4}$

$C=-1$

$4A=1 \rightarrow A=\frac{1}{4}$

$$\frac{1}{4}\int \frac{1}{x}\,dx + \int \frac{\frac{7}{4}x-1}{x^2+4}\,dx$$
$$\frac{1}{4}\ln|x| + \int \frac{\frac{7}{4}x}{x^2+4}\,dx - \int \frac{1}{x^2+4}\,dx$$
$$\frac{1}{4}\ln|x| + \frac{7}{4}\int \frac{x}{x^2+4}\,dx - \int \frac{1}{x^2+4}\,dx$$
$$\frac{1}{4}\ln|x| + \frac{7}{4}\int \frac{x}{(x^2+4)}\,dx - \frac{1}{2}\arctan\left(\frac{x}{2}\right)$$
$$\frac{1}{4}\ln|x| + \frac{7}{8}\ln|x^2+4| - \frac{1}{2}\arctan\left(\frac{x}{2}\right)+C$$

$x(x^2+4)^{-1} \rightarrow u^{-1}$

$u=x^2+4 \rightarrow u-4=x^2 \rightarrow \sqrt{u-4}=x$

$\frac{du}{dx}=2x$

$du=2x\,dx$

$\frac{du}{2\sqrt{u-4}}=dx$

$$\int \sqrt{u-4}\cdot u^{-1}\cdot \frac{du}{2\sqrt{u-4}}$$

$$\int \frac{\sqrt{u-4}\cdot u^{-1}}{2\sqrt{u-4}}\,du$$

$$\frac{1}{2}\int u^{-1}\,du$$

$$\frac{1}{2}\ln|u|$$

$$\frac{1}{2}\ln|x^2+4|$$

Ex 15.5

$$\int \frac{1}{x(x^2+1)^2}\,dx$$

$$\frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$A(x^2+1)^2 + (Bx+C)(x)(x^2+1) + (Dx+E)(x)$$
$$A(x^4+2x^2+1) + Bx^4 + Cx^3 + Bx^2 + Cx + Dx^2 + Ex$$
$$(A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A$$

$0x^4+0x^3+0x^2+0x+1$

$A+B=0 \quad A=1$

$C=0 \quad C=0$

$2A+B+D=0 \quad B=-1$

$C+E=0 \quad E=0$

$A=1 \quad D=-1$

$$\therefore \int \frac{1}{x(x^2+1)^2}\,dx = \int \frac{1}{x}\,dx + \int \frac{-x}{x^2+1}\,dx + \int \frac{-x}{(x^2+1)^2}\,dx$$
$$= \ln|x| - \int \frac{x}{u}\cdot\frac{du}{2x} - \int \frac{x}{u^2}\cdot\frac{du}{2x}$$
$$= \ln|x| - \frac{1}{2}\int \frac{1}{u}\,du - \frac{1}{2}\int \frac{1}{u^2}\,du$$
$$= \ln|x| - \frac{1}{2}\ln(x^2+1) + \frac{1}{2(x^2+1)} + C$$

Lec 16 - Integral approximations

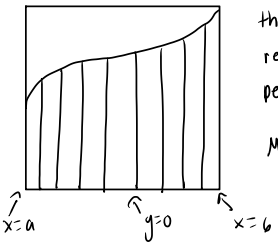
Problem

Sometimes we encounter a fun whose integral we cannot express with a simple formula.

e.g. $\int_a^b e^{-x^2}\,dx$ cannot be integrated according to our techniques. In such cases, it may be better to approximate an answer. Engineering sol'n cut out the shape of the shape of the area in a material with a known density, then weigh the cutout.

cs sol'n. try a monte carlo simulation

↳ $y=f(x)$

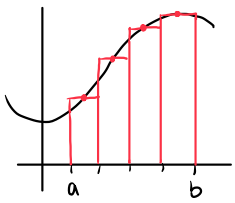


choose a point randomly in a rectangle that contains the area of interest. repeat 100 million times and record the percentage that fall inside the area.

Math sol'n: draw rectangles some more.

Solution

Riemann sums



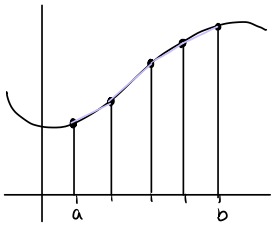
midpoint rule

$$M_n = \sum_{R=1}^{n} \Delta x \cdot f(a + (R-\tfrac{1}{2})\Delta x)$$

where $\Delta x = \frac{b-a}{n}$

now we could try approximating with more sophisticated shapes.

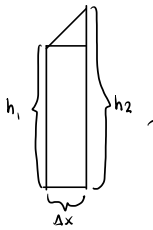
trapezoidal rule: draw trapezoids instead of rectangles.



Let $\Delta x = \frac{b-a}{n}$ and $x_R = a + R\Delta x$ for $R = 0, 1, \dots, n$.

The area of a trapezoid in the interval $[x_R, x_{R+1}]$ is

$$\Delta x \left(\frac{f(x_R) + f(x_R+1)}{2} \right)$$



$$\rightsquigarrow A = \Delta x\, h_1 + \frac{\Delta x \cdot (h_2 - h_1)}{2} = \Delta x \left(\frac{h_1}{2} + \frac{h_2}{2} \right)$$

So when we add up a series of these, we get

$$\Delta x \left(\frac{f(x_0)+f(x_1)}{2} \right) + \Delta x \left(\frac{f(x_1)+f(x_2)}{2} \right) + \dots + \Delta x \left(\frac{f(x_{R-1})+f(x_R)}{2} \right)$$

Trapezoidal Rule:

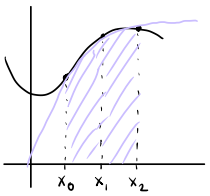
$$\int_a^b f(x)\,dx \approx T_n = \Delta x \left(\frac{f(a)}{2} + \sum_{R=1} f(a + R\Delta x) + \frac{f(b)}{2} \right)$$

where $\Delta x = \frac{b-a}{n}$.

Simpson's Rule

we can get an even better approximation if we use curved shapes (whose area we can compute).

Note: any three points on a plane uniquely identify a parabola. If we use three points from the boundaries of the two adjacent sub intervals, we can make a parabola.



Now each pair of subintervals will make a parabola (instead of one shape per sub interval), so we need an even number of sub intervals.

It is possible to compute the area under each parabola without directly finding the equations of the parabolas.

Area under a parabola that covers the sub interval $[x_n, x_{n+2}]$ (i.e. 2 regular subintervals)

$$\frac{\Delta x}{3} \left(f(x_R) + 4 f(x_{R+1}) + f(x_{R+2}) \right)$$

When we add all these up on $[a,b]$ with n subintervals (n is even), we get $\frac{\Delta x}{3}(f(x_0) + 4f(x_1) + f(x_2)) + \frac{\Delta x}{3}(f(x_2) + 4f(x_3) + f(x_4)) + \dots +$

$$\frac{\Delta x}{3} \left(f(x_{n-2}) + 4 f(x_{n-1}) + f(x_n) \right)$$

Simpson's rule:

$$\int_a^b f(x)\,dx \approx S_n =$$

$$\frac{\Delta x}{3} \left(f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n) \right)$$

where $\Delta x = \frac{b-a}{n}$ and $x_R = a + R\Delta x$ for $R = 0, 1, \dots, n$ and n is even.

Error in approximations

No approximation technique can be perfect, so there's bound to be error.

For example, if we're using the trapezoidal rule, the error $E_{T_n} = \left| \int_a^b f(x)dx - T_n \right|$.

the error from T_n depends on the concavity of f if f is very straight, then trapezoids do a good job. In general, if we know $|f''(x)| \le K_2$ on $[a,b]$, then $E_{T_n} \le \frac{K_2 (b-a)^3}{12 n^2}$. The error for the midpoint rule is $E_{M_n} \le \frac{K_2 (b-a)^3}{24 n^2}$.

Ex 16.1

$T_5 \approx \int_1^2 \frac{1}{x}\,dx$ → estimate

Let $f(x) = \frac{1}{x}$, $a=1$, $b=2$

we have $f'(x) = \frac{-1}{x^2}$, $f''(x) = \frac{2}{x^3}$

on $[1,2]$, we have $|f''(x)| \le 2$, so we take $K_2 = 2$ then the error is bounded by $E_{T_5} \le \frac{2(2-1)^3}{12 \cdot 5^2} = \frac{2}{300} \approx 0.006666\dots$

Ex 16.2

T_5 for $\int_1^2 \frac{1}{x}\,dx$ → actual error

$f'(x) \to f''(x) \to f'''(x) \to P$

$f^{(3)}(x)$

The error bound using Simpson's Rule is much better. Suppose $|f^{(4)}(x)| \le K_4$ on $[a,b]$. then the error E_{S_n} from Simpson's Rule is $E_{S_n} \le \frac{K_4 (b-a)^5}{180 \cdot n^4}$.

Ex 16.4

How many subintervals would we have to use in order for the error from Simpson's Rule in estimating $\int_1^2 \frac{1}{x}\,dx$ to be less than 0.0001?

we have $f''(x) = \frac{2}{x^3}$, $f^{(3)}(x) = \frac{-6}{x^4}$, $f^{(4)}(x) = \frac{24}{x^5}$, so on $[1,2]$, we have $|f^{(4)}(x)| \le 24$.

Then $E_{S_n} \le \frac{24(2-1)^5}{180 \cdot n^4}$ and we would like this to be less than $\frac{1}{10\,000}$

solve $\frac{24}{180\,n^4} \le \frac{1}{10\,000}$ $\Rightarrow$ $\frac{24\,000}{180} \le n^4$ $n \ge \sqrt[4]{\frac{4000}{30}} \approx 6.04\dots$

Lec 17 - Extreme values

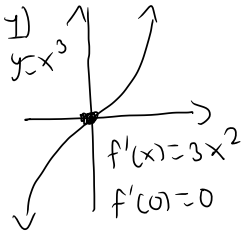
the extrema of a graph/fun are the points where the y-values are higher or lower than those around them.

global max/min: min or max value in all domain

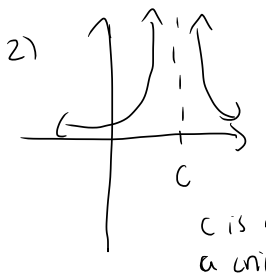
local min/max: min or max value in its surroundings

Critical points:

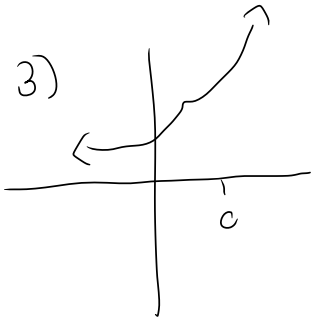
- $f'(c)=0$ (Type 1) ⇒ $f(c)$ exists
- or $f'(c)=DNE$ (Type 2) ↙ but $f(c)$ doesn't



critical number



c is not a critical number because its not in the domain



c is a type 2 critical number because $f'(c)$ DNE.

Extreme

sin

~~sec~~

tan

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}$$

$\sqrt{1-x^2}$ → $\sin\theta$

$\sqrt{1+x^2}$ → $\tan\theta$

$\sqrt{x^2-1}$ → $\sec\theta$

Reimann sum

$$\lim_{x\to\infty} \Delta x \sum_{i=0}^{n-1} f(a+\Delta x \cdot i)$$

$$\Delta x = \frac{b-a}{n} \qquad x_i = a + \Delta x \cdot i$$